1 Random For Loops

Problem 1.1. Consider the following algorithm,

(i) Determine the **best case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(ii) Determine the **worst case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(iii) Determine the expected case running time of the algorithm. **Fully justify your solution.**

Assume that Random(n) takes constant time and returns a number between 1 and n uniformly randomly.

```
function Func(A[],n)
    x = 0
    k = Random(⌊n/3⌋)
    for i = 1 to k do
        for j = 1 to i^2 do
            x = x + A[(ij mod n)]
        end for
    end for
    return x
end function
```

$$\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6} \in \Theta(n^2)$$

BC: Occurs when $k=1$, $B(n) \in \Theta(1)$

WC: occurs when $k = \frac{n}{3}$, $W(n) \in \Theta(n^3)$

EC:
$$ET(n) = \sum_{q=1}^{n/3} Pr(k=q)\,t(k=q)$$
$$= \sum_{q=1}^{n/3} 1/n/3 \cdot cq^3$$
$$= \frac{3c}{n} \sum_{q=1}^{n/3} q^3$$
$$= \frac{3c}{n} \cdot \frac{(n/3)^2\left(\frac{n}{3}+1\right)^2}{4}$$

Thus, $ET(n) \in \Theta(n^3)$.

Problem 1.2. Consider the following algorithm,

(i) Determine the **best case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(ii) Determine the **worst case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(iii) Determine the expected case running time of the algorithm. **Fully justify your solution.**

Assume that Random(n) takes constant time and returns a number between 1 and n uniformly randomly.

```
function Func(A[],n)
    x = 0
    for i = 1 to n do
        k = Random(⌊√i⌋)
        for j = 1 to k^2 do
            x = x + A[(ij mod n)]
        end for
    end for
    return x
end function
```

X_i

$X_i = ck^2$

$E(X_i) = \sum_{k=1}^{\sqrt{i}} \frac{1}{\sqrt{i}} ck^2$

$= \frac{c}{\sqrt{i}} \cdot \frac{\sqrt{i}(\sqrt{i}+1)(2\sqrt{i}+1)}{6}$

So, $E(X_i) \in \Theta(i)$

BC: Occurs when $k=1$ every iteration, $B(n) = \sum_{i=1}^{n} c \Rightarrow B(n) \in \Theta(n)$

BC: Occurs when $k=\sqrt{i}$ every iteration, $W(n) = \sum_{i=1}^{n} ci \Rightarrow W(n) \in \Theta(n^2)$

EC: $ET(n) = E\left[\sum_{i=1}^{n} X_i\right]$

$= \sum_{i=1}^{n} E[X_i]$

$= \sum_{i=1}^{n} ci$

$= c\frac{n(n+1)}{2}$

Thus, $ET(n) \in \Theta(n^2)$.

Problem 1.3. Consider the following algorithm,

(i) Determine the **best case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(ii) Determine the **worst case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(iii) Determine the expected case running time of the algorithm. **Fully justify your solution.**

Assume that $\texttt{Random}(i, j)$ returns a number between i and j uniformly randomly (including both endpoints).
Hint: If you are stuck, try creating a table listing all the possible outcomes, their associated probabilities, and their running times,

Event	Time	Probability

```
function Func(A[],i,j)
    z = 0
    n = j - i + 1
    k = Random(i, j)
    for x = i to k do
        for y = i to j do
            z = z + A[(xy mod n) + i]
        end for
    end for
    return x
end function
```

$X = (j-i+1)(k-i+1)c = cn(k-i+1)$

i) BC occurs when $k=i$ and takes $\Theta(n)$ time.

ii) WC occurs when $k=j$ and takes $\Theta(n^2)$ time.

iii)

Event	Time	Prob.
$k=i$	$cn(1)$	$1/n$
$k=i+1$	$cn(2)$	$1/n$
$k=i+2$	$cn(3)$	$1/n$
$\vdots$	$\vdots$	$\vdots$
$k=j-1$	$cn(n-1)$	$1/n$
$k=j$	$cn(n)$	$1/n$

$$ET(n) = \sum_{q=1}^{n} cn(q) \cdot 1/n$$
$$= c \sum_{q=1}^{n} q$$
$$= c \cdot \frac{n(n+1)}{2}$$

Thus, $ET(n) \in \Theta(n^2)$.

OR:

$$ET(n) = \sum_{q=i}^{j} cn(q-i+1) \cdot 1/n = \cdots \in \Theta(n^2).$$

2 Random If Statements

Problem 2.1. Consider the following algorithm,

(i) Determine the **best case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(ii) Determine the **worst case running time** of the algorithm and **state what condition must be true** for it to occur. No justification is needed!

(iii) Determine the expected case running time of the algorithm. **Fully justify your solution.**

Assume that $\texttt{Random}(n)$ takes constant time and returns a number between 1 and n uniformly randomly.

```
function Func(A[],n)
    x = 0
    k = Random(n^2)
    if k ≤ n then
        x = x + Foo(A[], n)
    else
        x = x + Bar(A[], n)
    end if
end function
```

Assume that $\texttt{Foo}(A[], n)$ takes $\Theta(n^2)$ time and $\texttt{Bar}(A[], n)$ takes $\Theta(n \log(n))$ time.

$$ET(n) = Pr(\text{if})\, t(\text{if}) + Pr(\text{else})\, t(\text{else})$$

$$= \frac{n}{n^2} \cdot (cn^2) + \left(1 - \frac{n}{n^2}\right) \cdot cn\log(n)$$

$$= cn + cn\log(n) - c\log(n)$$

So, $ET(n) \in \Theta(n\log(n))$.

Problem 2.2. Consider the following algorithms,

(i) Express the best case running time of the algorithm as a recurrence relation and state what condition(s) must be true for it to occur.

(ii) Express the worst case running time of the algorithm as a recurrence relation and state what condition(s) must be true for it to occur.

(iii) Express the expected case running time of the algorithm as a recurrence relation

Assume that $\texttt{Random}(n)$ takes constant time and returns a number between 1 and n uniformly randomly.
YOU DO NOT NEED TO ANALYZE THE RECURRENCES. JUST WRITE THEM DOWN.

(a)

```
function Func(A[],n)
    if n ≤ 25 then
        return x
    end if
    for i = 1 to n do
        x = x + A[i]
    end for
    k = Random(n)
    if k ≤ ⌊3n/4⌋ then
        x = x + Func(A[], n − 2)
    else
        x = x + Func(A[], n/8)
    end if
    return x
end function
```

i) BC occurs when $k > 3n/4$ ALWAYS.

$B(n) = cn + B(n/8)$

ii) WC occurs when $k \leq 3n/4$ ALWAYS.

$W(n) = cn + W(n-2)$

iii) $ET(n) = cn + \frac{3}{4} ET(n-2) + \frac{1}{4} ET(n/8)$

(b)

```
function Func(A[],n)
    if n ≤ 25 then
        return x
    end if
    for i = 1 to n^2 do
        x = x + A[i]
    end for
    k = Random(n)
    if k mod 5 = 2 then
        for i = 1 to 6 do
            x = x + Func(A[], n/3)
        end for
    else
        x = x + Func(A[], n − 2)
        x = x + Func(A[], n − 6)
    end if
    return x
end function
```

i) WC occurs when $k \bmod 5 \neq 2$ ALWAYS.

$W(n) = cn^2 + W(n-2) + W(n-6)$

ii) BC occurs when $k \bmod 5 = 2$ ALWAYS.

$B(n) = cn^2 + 6B(n/3)$

iii) $ET(n) = cn^2 + \frac{4}{5}\left[ET(n-2) + ET(n-6)\right] + \frac{1}{5}\left[6\,ET(n/3)\right]$

Problem 2.3. Determine the asymptotic complexity of each of the following recurrence relations. Justify your answer with either substitution, a recursion tree, or induction. If the problem takes at least exponential time, then simply find exponential lower bounds.

(a)

$$T(n) = cn^2 + \frac{1}{6}T(n-4), \qquad T(0) = c$$

FILTHY trickery: $T(n) \le cn^2 + \frac{1}{6}T(n)$; $T(n) \ge cn^2$

$$\frac{5}{6}T(n) \le cn^2$$

$$T(n) \le \frac{6}{5}cn^2$$

More normal way:

$$T(n) = cn^2 + \frac{1}{6}\left[c(n-4)^2 + \frac{1}{6}T(n-8)\right]$$

$$T(n) \le cn^2 + \frac{1}{6}c(n-4)^2 + \frac{1}{6^2}T(n-8)$$

$$T(n) \le cn^2 + \frac{1}{6}c(n-4)^2 + \frac{1}{6^2}c(n-8)^2 + \frac{1}{6^3}T(n-12)$$

In general,

$$T(n) = \sum_{i=1}^{k-1} \frac{1}{6^i} c(n-4i)^2 + \frac{1}{6^k}T(n-4k)$$

Choose k s.t.

$n - 4k = 0$

$k = n/4$

$$T(n) = \sum_{i=1}^{k-1} \frac{1}{6^i} c(n-4i)^2 + \frac{1}{6^{n/4}}c$$

LB: $T(n) \ge cn^2$

UB: $T(n) \le \sum_{i=1}^{k-1} \frac{1}{6^i} cn^2 + \frac{c}{6^{n/4}}$

$T(n) \le cn^2 \sum_{i=1}^{\infty} (1/6)^i + \frac{c}{6^{n/4}}$

$T(n) \le cn^2 \left[\frac{1}{1-1/6}\right] + \frac{c}{6^{n/4}}$

$O(n^2)$

Thus, $T(n) \in \Theta(n^2)$.

LB: $T(n) \geq cn^2$.

(b)

$$T(n) = cn^2 + T(n/2) + \frac{6}{7}T(n/4), \qquad T(5) = c$$

Hint: Consider bounding the problem as the very first step.

2 common upper bounds: $T(n) \leq cn^2 + T(n/2) + \frac{6}{7}T(n/2)$
then use substitution

OR $T(n) \leq cn^2 + T(n/2) + T(n/4)$
then use tree.

I will use a 3rd option.

$$T(n) \leq cn^2 + T(n/2) + T(n/2)$$

$$T(n) \leq cn^2 + 2T(n/2)$$

$$T(n) \leq cn^2 + 2c\left(\frac{n}{2}\right)^2 + 4T(n/4)$$

$$T(n) \leq cn^2 + 2c\left(\frac{n}{2}\right)^2 + 4c\left(\frac{n}{4}\right)^2 + 8T(n/8)$$

In general,

$$T(n) \leq \sum_{i=0}^{k-1} 2^i c \left(\frac{n}{2^i}\right)^2 + 2^k T(n/2^k)$$

$$T(n) \leq cn^2 \underbrace{\sum_{i=0}^{k-1} (1/2)^i}_{\text{geo w/ } r<1 \text{ so } \Theta(1)} + \underbrace{2^{\log_2(n/5)} c}_{\Theta(n)}$$

Choose k s.t.

$$\frac{n}{2^k} = 5$$

$$k = \log_2(n/5)$$

Thus, $T(n) \in O(n^2)$.

So, $T(n) \in \Theta(n^2)$.

Problem 2.4. Consider the following algorithm,

(i) Determine the best case running time of the algorithm.

(ii) Determine the worst case running time of the algorithm.

(iii) Determine the expected case running time of the algorithm. **Fully justify your solution.**

```
function Func(A[],n)
    x = 0
    for i = 1 to n^2 do
        x = x + Foo(A[], i, n^2)
    end for
end function
```

$\Theta(1)$

$\Theta(\sqrt{n^2-i})$

$\Theta(n^2-i)$

where $\texttt{Foo}(A[], i, j)$ has the following properties (assuming $n = j - i + 1$),

- The best case running time $\Theta(1)$
- The expected case running time is $\Theta(\sqrt{n})$
- The worst case running time is $\Theta(n)$

BC: $B(n) = \sum_{i=1}^{n^2} c = cn^2$ So, $B(n) \in \Theta(n^2)$.

EC: $ET(n) = \sum_{i=1}^{n^2} c\sqrt{n^2-i} \le \sum_{i=1}^{n^2} c\sqrt{n^2} = n^2 \cdot cn$

$\ge \sum_{i=1}^{n^2/2} c\sqrt{n^2 - n^2/2} = \frac{n^2}{2} \cdot c\sqrt{\frac{n^2}{2}}$

So, $ET(n) \in \Theta(n^3)$

WC: $W(n) = \sum_{i=1}^{n^2} c(n^2 - i) = cn^4 - c\frac{n^2(n^2+1)}{2}$ So, $W(n) \in \Theta(n^4)$